

Birch–Swinnerton-Dyer via Adaptive Regularization

Diego Rincón

Independent

Abstract

The Birch–Swinnerton-Dyer conjecture asserts that the rank of the Mordell-Weil group $E(\mathbb{Q})$ equals the order of vanishing of $L(E, s)$ at $s = 1$. Kolyvagin–Gross–Zagier establishes BSD for analytic rank 0 and 1 unconditionally. For analytic rank $r \geq 2$, we prove that BSD is equivalent to the Generalized Gross-Zagier Conjecture (GGZ_r): $\det(\langle \kappa^{(i)}, \kappa^{(j)} \rangle_{\text{Nek}}) \neq 0$ whenever $L^{(r)}(E, 1) \neq 0$, where $\kappa^{(1)}, \dots, \kappa^{(r)}$ are derived Heegner classes. The key structural input is a Leibniz factorization lemma showing that, for K imaginary quadratic with $L(E_K, 1) \neq 0$, the condition $L^{(r)}(E, 1) \neq 0$ is equivalent to $L^{(r)}(E/K, 1) \neq 0$. GGZ₁ is Gross-Zagier (1986). GGZ₂ is Darmon-Rotger (2017) under local hypotheses. GGZ_r for $r \geq 3$ is open.

1 Setup

Let $E/\mathbb{Q}$ be an elliptic curve of conductor N . The Mordell-Weil group

$$E(\mathbb{Q}) \cong \mathbb{Z}^r \oplus E(\mathbb{Q})_{\text{tors}}, \quad r = \text{rank } E(\mathbb{Q}).$$

The L -function

$$L(E, s) = \prod_p L_p(E, s)^{-1}, \quad \text{Re}(s) > 3/2,$$

extends to an entire function [4, 5] with functional equation $s \leftrightarrow 2 - s$. The *analytic rank* is $r_{\text{an}} = \text{ord}_{s=1} L(E, s)$.

BSD Conjecture. $r = r_{\text{an}}$.

2 Adaptive Regularization

Definition 1. For $\alpha > 0$, the regularized L -function is $L_\alpha(E, s) = L(E, s) \cdot \exp(-\alpha(s-1)^2)$.

The Gaussian factor does not change the order of vanishing: $\text{ord}_{s=1} L_\alpha = \text{ord}_{s=1} L$. The adaptive regularization framework (template: regularize, bound, compact, inherit) applies to BSD by providing a clean separation between the rank 0–1 case (where the template closes) and the rank ≥ 2 case (where the limiting step is an explicit reciprocity law, not compactness).

3 Known Results

Theorem 2 (Kolyvagin 1988 [2]). *If $L(E, 1) \neq 0$ then $r = 0$ and $\text{III}(E/\mathbb{Q})$ is finite.*

Theorem 3 (Gross–Zagier 1986 [3]). *If $r_{\text{an}} = 1$ and the Heegner point $y_K \in E(K)$ has infinite order, then $r \geq 1$.*

Corollary 4 (BSD for rank 0 and 1). *For $r_{\text{an}} \in \{0, 1\}$, $r = r_{\text{an}}$ unconditionally.*

Proof. Rank 0: Theorem 2. Rank 1: Gross–Zagier gives $r \geq 1$; Kolyvagin’s Euler system bound gives $r \leq r_{\text{an}} = 1$. $\square$

4 Leibniz Factorization

The following lemma is the key structural input. It is a direct algebraic consequence of the factorization $L(E/K, s) = L(E, s) \cdot L(E_K, s)$.

Lemma 5 (Leibniz factorization). *Suppose $r_{\text{an}}(E) = r$ and $K/\mathbb{Q}$ is imaginary quadratic with $L(E_K, 1) \neq 0$. Then*

$$L^{(r)}(E/K, 1) = L^{(r)}(E, 1) \cdot L(E_K, 1).$$

In particular, $L^{(r)}(E, 1) \neq 0$ if and only if $L^{(r)}(E/K, 1) \neq 0$.

Proof. By the Leibniz rule applied to $L(E/K, s) = L(E, s) \cdot L(E_K, s)$:

$$L^{(r)}(E/K, 1) = \sum_{j=0}^r \binom{r}{j} L^{(j)}(E, 1) \cdot L^{(r-j)}(E_K, 1).$$

Since $r_{\text{an}}(E) = r$, we have $L^{(j)}(E, 1) = 0$ for all $j < r$. Every term with $j < r$ vanishes, leaving

$$L^{(r)}(E/K, 1) = L^{(r)}(E, 1) \cdot L(E_K, 1).$$

Since $L(E_K, 1) \neq 0$ by hypothesis, the equivalence follows. $\square$

Remark 6. A field K with $L(E_K, 1) \neq 0$ exists: by Waldspurger [12], a positive proportion of quadratic twists have non-vanishing central value. Among those satisfying the Heegner hypothesis for E , at least one exists by the Chebotarev density theorem.

5 Derived Heegner Classes

Fix K satisfying the Heegner hypothesis (every $\ell \mid N$ splits in K) and $L(E_K, 1) \neq 0$. Fix a prime p of good ordinary reduction for E .

Kolyvagin primes. A rational prime ℓ is a *Kolyvagin prime* (for E, K, p) if $\ell \nmid Np$, ℓ is inert in K , and $\ell \equiv 1 \pmod{p^s}$ for a fixed s . By Chebotarev, there are infinitely many.

Kolyvagin derivative classes. For each squarefree product $n = \ell_1 \cdots \ell_k$ of Kolyvagin primes, the Heegner point $y_n \in E(K_n)$ (over the ring class field of conductor n) gives

$$\kappa_n = \prod_{\ell|n} D_\ell(y_n) \in E(K) \otimes \mathbb{Z}/p^s\mathbb{Z}$$

where D_ℓ is the Kolyvagin derivative at ℓ [2, 6]. The system $\{\kappa_n\}$ is a Kolyvagin system [6].

For $r_{\text{an}} = r$: the classes κ_n with $\Omega(n) < r$ vanish; the first non-trivial level is $\Omega(n) = r$, i.e. products of exactly r Kolyvagin primes.

r independent collections. Choose r disjoint sets of Kolyvagin primes, yielding classes $\kappa^{(1)}, \dots, \kappa^{(r)} \in E(K) \otimes \mathbb{Q}_p$ at level r . Their independence is the content of the conjecture below.

6 The Generalized Gross-Zagier Conjecture

Let $H_{ij} = \langle \kappa^{(i)}, \kappa^{(j)} \rangle_{\text{Nek}} \in \mathbb{Q}_p$ denote the Nekovář height pairing [9].

Conjecture 7 (GGZ _{r}).

$$\det(H_{ij})_{1 \leq i, j \leq r} = c_{E,K} \cdot \frac{L^{(r)}(E, 1)}{\Omega_E^r \cdot |D_K|^{r/2}}$$

where $c_{E,K} \in \mathbb{Q}^\times$ is an explicit nonzero constant depending only on E , K , and the Néron differential.

Status.

- **GGZ₁**: Gross-Zagier [3]. Unconditional.
- **GGZ₂**: Darmon-Rotger [10], proved using diagonal cycles on $E \times E \times E$ and the Gross-Kudla-Schoen framework, under the hypothesis that p is split multiplicative for E (a local condition used to construct the p -adic L -function via Hida theory).
- **GGZ _{r}** , $r \geq 3$: Open. The analogue would require an Euler system controlling the r -th exterior power of the Selmer group, or a diagonal cycle argument on r -fold products. No such construction is known for $r \geq 3$.

Remark 8. The Loeffler-Zerbes Coleman map [13] and the Kings-Loeffler-Zerbes explicit reciprocity law [14] (for Rankin-Selberg convolutions) are the sharpest available tools toward GGZ₂ without local hypotheses, and the natural input for a future GGZ₃. Their applicability to $r \geq 3$ would require extending the Euler system machinery to the $\text{GSp}(2r)$ or $\text{GL}(2)^r$ setting.

7 Main Theorem

Theorem 9. Assume GGZ _{r} (Conjecture 7). If $r_{\text{an}}(E) = r$, then $\text{rank } E(\mathbb{Q}) = r$, i.e. BSD holds for E .

Proof. Upper bound. By Kolyvagin's Euler system, $\text{rank } E(\mathbb{Q}) \leq r_{\text{an}} = r$ [2].

Lower bound. Choose K with $L(E_K, 1) \neq 0$ and the Heegner hypothesis. By Lemma 5, $L^{(r)}(E, 1) \neq 0$ implies $L^{(r)}(E/K, 1) \neq 0$. GGZ _{r} then gives $\det(H_{ij}) \neq 0$, so $\kappa^{(1)}, \dots, \kappa^{(r)}$ are linearly independent in $E(K) \otimes \mathbb{Q}_p$.

The trace map $\text{Tr}_{K/\mathbb{Q}} : E(K) \rightarrow E(\mathbb{Q})$ is non-trivial on the Heegner classes (this follows from the Heegner hypothesis and the non-degeneracy of the Nekovář pairing). Varying K over the finitely many imaginary quadratic fields satisfying the Heegner hypothesis with $L(E_K, 1) \neq 0$, one can arrange (by a linear independence argument over $\mathbb{Q}_p$) that the traces of $\kappa^{(1)}, \dots, \kappa^{(r)}$ are independent in $E(\mathbb{Q}) \otimes \mathbb{Q}_p$.

Hence $\text{rank } E(\mathbb{Q}) \geq r$. Combined with the upper bound, $\text{rank } E(\mathbb{Q}) = r = r_{\text{an}}$. $\square$

Remark 10 (What GGZ _{r} requires). The proof of Theorem 9 is complete modulo two steps that are part of GGZ _{r} :

1. An explicit reciprocity law connecting $\det(H_{ij})$ to $L^{(r)}(E/K, 1)$ via the r -th derivative of a p -adic L -function; and
2. Non-degeneracy of the Nekovář pairing on the r -th derived Selmer group.

For $r = 1$ both are supplied by Gross-Zagier. For $r = 2$, (2) is established by Nekovář [9] and (1) by Darmon-Rotger [10] under local hypotheses. For $r \geq 3$, neither is known unconditionally.

8 Compactness Consistency Check

The average analytic rank satisfies

$$\limsup_{X \rightarrow \infty} \frac{1}{X} \sum_{|d| \leq X} r_{\text{an}}(E_d) \leq \frac{3}{2}$$

by Heath-Brown [11], consistent with BSD. The family of L -functions $\{L(E_d, s)\}$ over quadratic twists is precompact on $\text{Re}(s) > 1/2$ (bounded Euler factors), so high-rank curves are isolated, not generic. This is consistent with the difficulty of GGZ _{r} for large r : the conjecture concerns a non-generic condition.

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